

Every value is $f^*(2\sigma)$ (smallest injected fraction detected at 2σ median significance; lower is better), median over draws. Residual children indented; best test per row **bold**. '>.1' = never reached 2σ .

method	Eucl.	Maha.	MMD	SparKer
SupCon	.043	.057	.020	.018
SupCon+SIGReg	>.1	>.1	.031	.037
NPLM-dist	.024	.038	.039	.027
NPLM+cwSIGReg	.022	.032	.033	.027
proto/MSE+cwSIGReg	.046	.059	.041	.039
SupCon+cwSIGReg	.066	.048	.033	.058
SupCon $\rightarrow$ res	.033	.023	.033	.022
SupCon $\rightarrow$ res-nplm	.033	.023	.029	.017
SupCon $\rightarrow$ res	>.1	.029	.044	.069
SupCon $\rightarrow$ res-nplm	.025	.044	.038	.022
NPLM-dist $\rightarrow$ res	.084	.041	.046	.046
NPLM-dist $\rightarrow$ res-nplm	>.1	.048	.044	.039
NPLM $\rightarrow$ res	>.1	.023	.037	.048
NPLM $\rightarrow$ res-nplm	.045	.034	.036	.043
proto/MSE $\rightarrow$ res	>.1	.062	.044	.039
proto/MSE $\rightarrow$ res-nplm	.058	.053	.043	.043

Table 1: CIFAR-10: pre-discovery (frozen) f^* , median over draws.

method	Eucl.	Eucl.(a)	Maha.	MMD	SparKer	SpK(a)
SupCon	.014	.013	.016	.025	.017	.014
SupCon+SIGReg	.037	.040	.064	.034	.037	.042
NPLM-dist	.045	.021	.037	.036	.032	.025
NPLM+cwSIGReg	.044	.015	.031	.044	.033	.020
proto/MSE+cwSIGReg	.037	.050	.043	.061	.037	.065
SupCon+cwSIGReg	.041	.030	.049	.035	.048	.050
SupCon $\rightarrow$ res	.017	.015	.016	.030	.017	.018
SupCon $\rightarrow$ res-nplm	.021	.018	.018	.029	.019	.019
SupCon $\rightarrow$ res	.053	.030	.030	.042	.051	.046
SupCon $\rightarrow$ res-nplm	.039	.017	.031	.032	.026	.019
NPLM-dist $\rightarrow$ res	.036	.030	.028	.040	.037	.032
NPLM-dist $\rightarrow$ res-nplm	.086	.033	.058	.050	.027	.045
NPLM $\rightarrow$ res	.040	.040	.023	.044	.033	.040
NPLM $\rightarrow$ res-nplm	.071	.040	.027	.043	.042	.040
proto/MSE $\rightarrow$ res	.030	.020	.022	.046	.039	.026
proto/MSE $\rightarrow$ res-nplm	.028	.023	.026	.046	.029	.029

Table 2: CIFAR-10: post-discovery f^* , derived cut (np pool). Eucl.(a)/SpK(a) = anchor-aware.

CIFAR-10 (10 classes, from-scratch)

method	Eucl.	Eucl.(a)	Maha.	MMD	SparKer	SpK(a)
SupCon	.022	.015	.019	.021	.017	.015
SupCon+SIGReg	.026	.023	.030	.035	.024	.022
NPLM-dist	.029	.019	.025	.041	.028	.034
NPLM+cwSIGReg	.022	.014	.022	.048	.025	.020
proto/MSE+cwSIGReg	.028	.025	.039	.035	.038	.035
SupCon+cwSIGReg	.031	.025	.027	.032	.037	.042
SupCon $\rightarrow$ res	.018	.017	.016	.031	.018	.018
SupCon $\rightarrow$ res-nplm	.021	.016	.018	.024	.021	.019
SupCon $\rightarrow$ res	.022	.018	.016	.039	.033	.039
SupCon $\rightarrow$ res-nplm	.017	.016	.022	.029	.021	.017
NPLM-dist $\rightarrow$ res	.020	.016	.018	.046	.026	.027
NPLM-dist $\rightarrow$ res-nplm	.053	.018	.046	.040	.043	.042
NPLM $\rightarrow$ res	.019	.017	.016	.040	.026	.026
NPLM $\rightarrow$ res-nplm	.022	.015	.019	.033	.031	.029
proto/MSE $\rightarrow$ res	.020	.022	.021	.051	.026	.024
proto/MSE $\rightarrow$ res-nplm	.028	.025	.027	.041	.028	.024

Table 3: CIFAR-10: post-discovery f^* , $s/\sqrt{b}$ cut.